

Self-Heating Effect of Device-Circuit with Back-side Power Delivery Network beyond 3nm Node

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Abstract—In this work, we explored the self-heating effect (SHE) and performance analysis of Backside Power Delivery Network (BSPDN) technology for routing logical signals from underlying devices to Static Random (SRAM) macros and 3nm technology nodes. A comprehensive evaluation was conducted on the impact of using BSPDN on the thermal effects and performance of circuits in terms of Power, Performance, Area, and Thermal (PPAT) from devices to circuits. And a thermal network model based on device physics is used to describe the SHE on the SRAM macro. This article studies and analyzes the thermal RC network model of the device circuit. Compared with the same design as traditional Front Side (FS) PDNs, the BSPDN structure has better performance, stability, and area advantages. However, owing to the distinctive structural characteristics, the co-design and optimization of circuit performance and thermal reliability using BSPDN present additional challenges.

Index Terms-- Self-Heating Effect (SHE), Backside Power Delivery Network (BSPDN), 3nm technology nodes, Thermal network model, Power Performance Area Thermal (PPAT).

I. INTRODUCTION

The collaborative optimization of device performance and design continues to extend the process nodes towards 2nm technology nodes, introducing solutions such as stacked gate-all-around (GAA) nanosheets (NSH) FETs [1,2]. And stacked gate-all-around (GAA) nanosheets (NSH) FETs are also being further scale down, converted into complementary transistors (CFET) [3,4]. However, the scalability to large-scale technology nodes has brought forth significant challenges in the design of both the Front-End-of-Line (FEOL) and Back-End-of-Line (BEOL) components, including issues related to pin access and wiring congestion [5-7]. The innovative technologies are imperative to address these bottlenecks and facilitate further expansion in both device and integrated circuit (IC) capabilities [8]. BSPDN technology leverages Backside Metal (BSM), Buried Power Rails (BPR), and nano Through-Silicon Vias (nTSV) to interconnect BSM and BPR, offering advantages in mitigating the IR voltage drop in both 2D [5] and 3D chips [5], while alleviating strain on the BEOL wiring [9,10]. Nonetheless, the thermal ramifications and potential performance degradation resulting from the BSPDN architecture remain a prominent concern. Therefore, SHE is imperative for bridging the gap between device-level considerations and circuit design [11].

By connecting nTSV [12] as a local buried power rail (BPR) to the back metal layer of PDN [13], the use of BPR

power rail can eliminate the VDD/VSS BEOL layer above FEOL [14-17], achieving height scaling of device standard units. In Fig. 1, we show a three-dimensional schematic of a 6T-SRAM standard circuit unit with a BSPDN structure. The middle part of the figure is an SRAM circuit consisting of six transistors under 3nm technology nodes. The front end is the signal routing of the circuit, and the back end is the buried power rail of the circuit. The FSPDN and BSPDN components for 3D IC circuits are shown in Fig. 2. Compared to the BSPDN structure, FSPDN is a traditional PDN structure and shares the BEOL with the signal. The local PDN network consists of BPR, M1, and M3, while the global network consists of M4 and M12. Above M12, integrated by 2D MIMCAP [18] (traditional metal insulator metal capacitor in BEOL) or 2.5D MIMCAP (a proposed high-density MIMCAP with additional vertical capacitors integrated in BEOL, with an unpacking density more than 6 times that of 2D cases [19,20]).

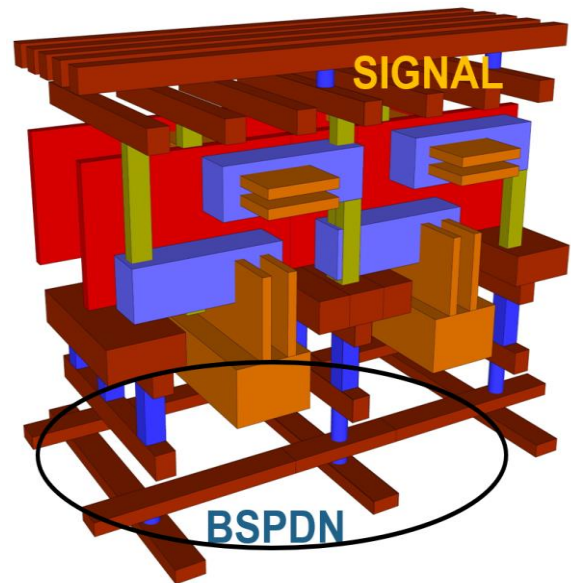


Figure 1. Back Side Power (BSP) scheme, using nTSV to connect Buried Power Rails (BPR)

Fig. 3 shows the layout structure of 6T-SRAM with BSPDN structure. Within this context, the distribution of circuit path temperature with different BPR materials in this structure was compared.

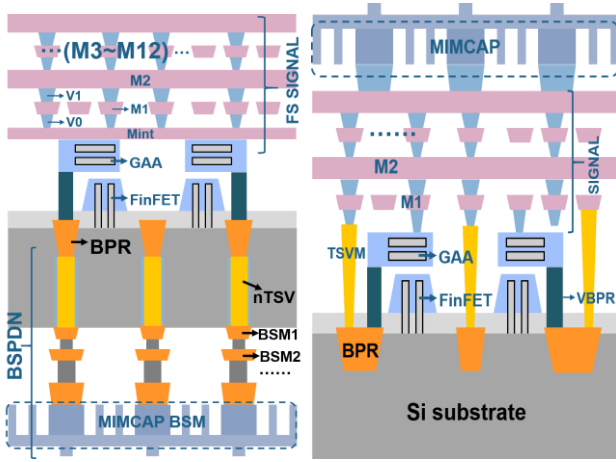


Figure 2. (a) Schematic of BSPDN (or (b), FSPDN), with MIMCAP integrated in the bottom (or top) metal layers

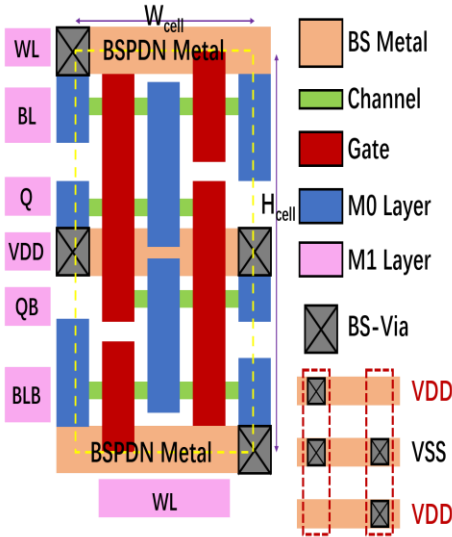


Figure 3. Layout of 6T-SRAM structure with BSPDN structure

II. THERMAL BEHAVIOR ANALYSIS

Based on the layout structure of 6T-SRAM in Fig. 3, the circuit path temperature distribution of different BPR materials in this structure was compared. Fig. 4 provides an in-depth understanding of the temperature distribution and temperature diffusion direction observed during the thermal simulation process. It can be seen that in the thermal simulation process of SRAM circuit, the device passes through the drain and gate at the hottest position in the circuit, then transfers to BPR, and finally transfers to PDN-BEOL. It can be seen that the thermal behavior of BSPDN structure is significantly different from that of FSPDN structure, and its thermal diffusion path is upward or downward diffusion.

Fig. 5 evaluates the dynamic response characteristics of BSPDN and FSPDN structures by selecting the hottest transistor in the 6T-SRAM circuit. Due to the higher volume thermal capacitance of Cu compared to oxide, it can lead to slower heating speed and lower delay. In Fig. 5, this thermal

delay can be seen starting from 10^{-11} s, where the transient response with and without BSPDN is compared.

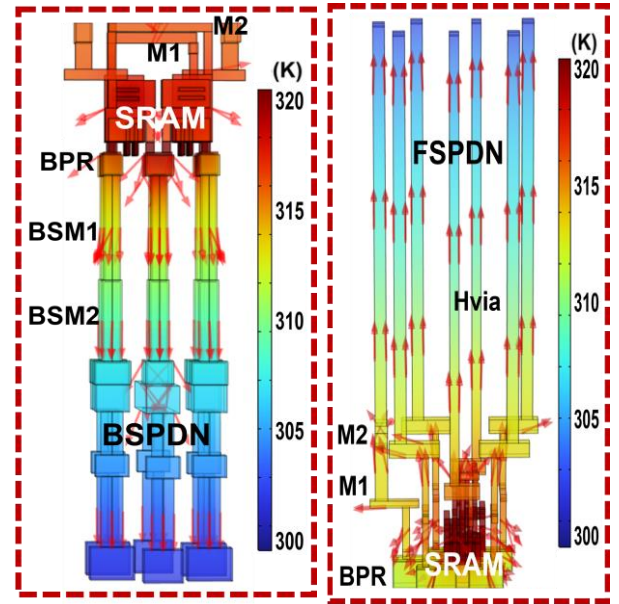


Figure 4. 3D view of temperature distribution in SRAM circuits through finite element simulation (BSPDN and FSPDN structure)

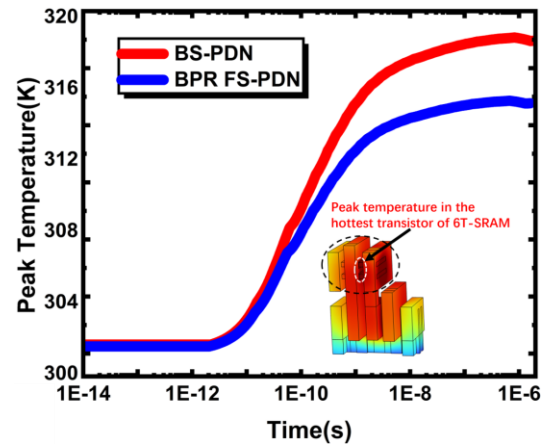


Figure 5. The transient thermal effect in the back PDN based on the thermal network model is selected as the peak temperature in the highest temperature transistor of 6T-SRAM with different structures.

III. RESULTS AND DISCUSSION

Meanwhile, when nTSV connects BPR and BSM together to form the BSPDN structure of the circuit, we analyzed the influence of nTSV radius size and height on temperature, as shown in Fig. 6. It is worth noting that the increase in nTSV radius and height directly corresponds to the rising temperature. And with the increase of TSV radius and TSV height, the temperature of the entire structure of BSPDN will increase.

Fig. 7 shows the temperature comparison between BSPDN and FSPDN structures with and without MIMCAP integration. It can be seen from the figure that compared to FSPDN

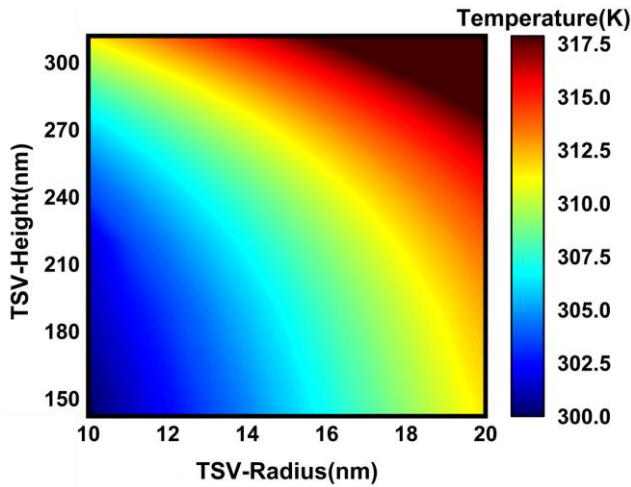


Figure 6. The impact of changes in TSV radius and height on circuit temperature, as the TSV structure changes, the peak temperature of the device becomes higher and higher.

structures, BSPDN has a higher overall temperature due to its high scaling ability. Regardless of the existence of MIMCAP structure, the temperature of BSPDN structure is higher than that of FSPDN structure, and the presence of MIMCAP structure will also exacerbate the heat dissipation of buried power rails. Compared to BSPDN with different circuit structures, the temperature of SRAM circuit is also much higher than that of NAND circuit.

For the BSPDN structure, due to its multi-layered metal layer interconnection structure, Fig. 8 investigates the temperature comparison of the heat flux distribution of different metal layers in BSPDN. From the graph, it can be seen that the heat flux distribution of both structures shows a trend of increasing with the height of the metal layer. The heat flux diffuses through BPR, TSV, and BSM, and reaches the maximum heat flux distribution when it reaches the second layer of the metal layer.

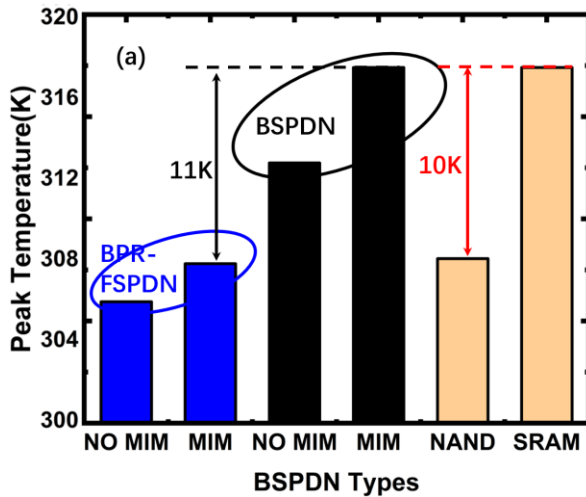


Figure 7. Temperature comparison of two structures with and without MIMCAP integration and temperature comparison of different circuits under BSPDN structure

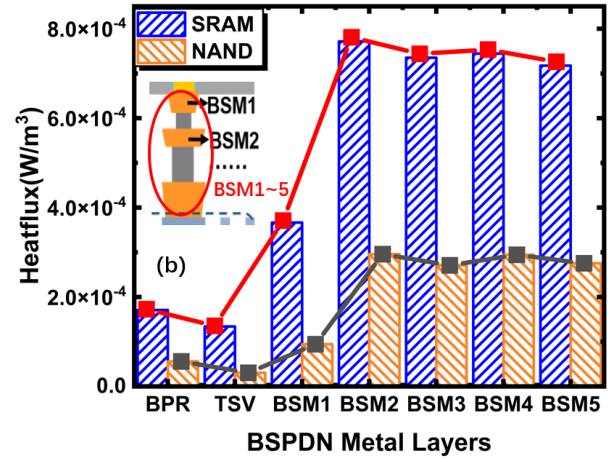


Figure 8. Heat flux distribution of each layer in BSPDN structure

Fig. 9 shows the temperature comparison of the effect of BSPDN material and number of layers on SHE. It can be seen that the number of BSM layers has a significant impact on the temperature of the BSPDN structure. The fewer BSM layers, the higher the temperature of the BSPDN structure. Due to the issue of thermal conductivity of different materials, it can be seen from the figure that Cu material should be selected as the main material for BSPDN interconnection structure, which is more suitable for heat dissipation, and the number of BSM layers should also be better between 4-5. Overall, the number of BSM layers and the use of different thermal conductivity materials in the BSPDN structure also have a significant impact on the overall circuit SHE.

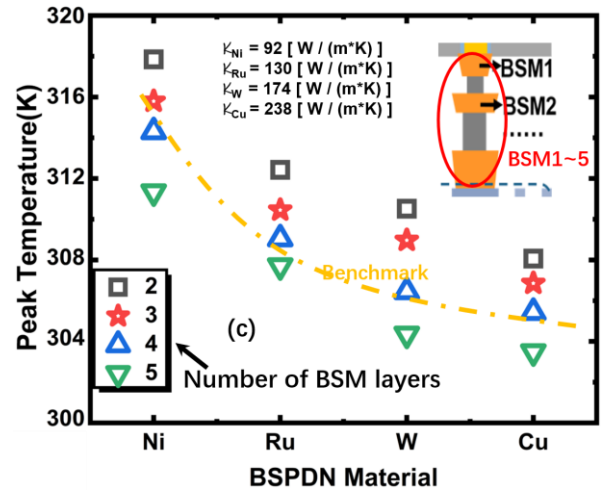


Figure 9. Effect of BSM material and number of BSM layers on circuit temperature

Within this context, the distribution of circuit path temperature distribution of different BPR materials in this structure was compared, as shown in Fig. 10. For SRAM circuits, different metal materials are used for analysis of different input signal circuits and power supply rails. Due to the different thermal conductivity of various materials and the different working states of SRAM wire and bit line, it can be seen that the highest temperature occurs when the wire is working. The temperature generated by using nickel material

is generally higher than other materials, although it can produce good electrical properties.

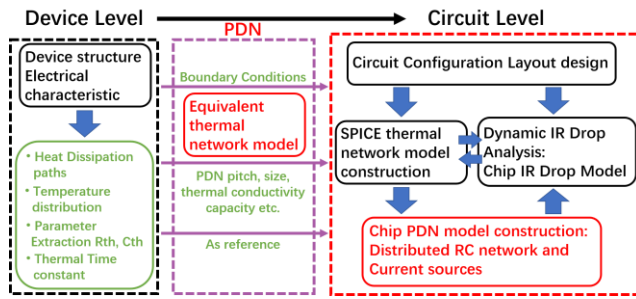


Figure 10. Framework of the Device-PDN-Circuit level self-heating and performance evaluation process. By simulating the electrical characteristics at the device level through TCAD, Rth and Cth are extracted to analyze the equivalent thermal network model of the device and PDN structure, and further analyze the impact of thermal performance degradation on circuit performance.

Based on all the research results related to heat, we propose a thermal behavior analysis framework for device level PDN circuits, which explains the influence of device level self-heating behavior on the temperature of BSPDN structures, as shown in Fig. 10. At the device level, it is mainly based on Technology Computer Aided Design (TCAD) simulation software to simulate the device structure and electrical characteristics, and obtain the thermal distribution path, heat flux distribution, and the extracted thermal time constant, thermal resistance, and heat capacity of the device.

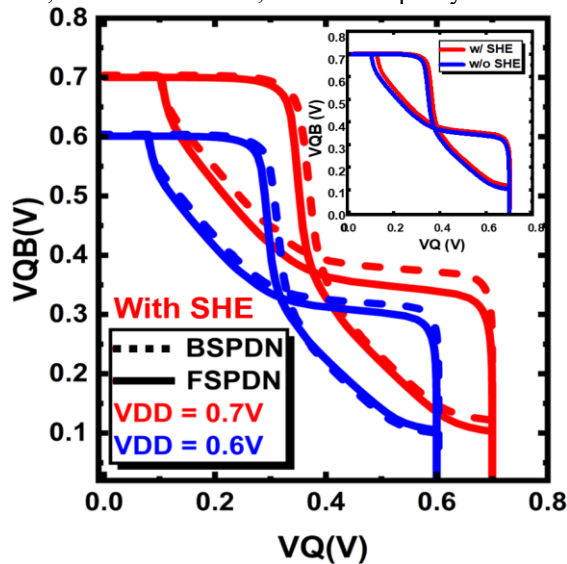


Figure 11. The noise tolerance of 6T-SRAM is determined by different PDN types (with and without SHE). Under the influence of self-heating, the static noise tolerance of 6T-SRAM under BSPDN structure is significantly different from that of FSPDN structure.

At the circuit level, the thermal distribution and electrical degradation at the circuit level are obtained through SPICE thermal network model analysis and dynamic IR drop analysis. However, using different PDN structures between devices and circuits can also have significant changes in the performance of the entire hierarchy. For the middle part, the thermal behavior evaluation and analysis of the PDN structure first determine the pitch, size, and thermal conductivity of related

materials by setting boundary conditions for the structure, and then evaluate it using an equivalent thermal conductivity model. The increasingly important self-heating behavior of devices and circuit reliability jointly promote serious degradation of circuit performance, and the BSPDN structure exacerbates the occurrence of self-heating behavior, making the degradation even more severe. Therefore, the framework of Device-PDN-Circuit level self-heating and performance evaluation process has very important guidance and reference significance for subsequent research.

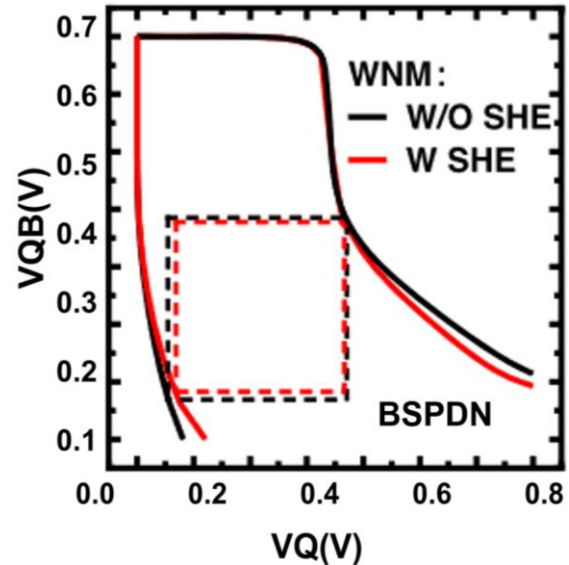


Figure 12. The noise tolerance of 6T-SRAM is determined by different PDN types (with and without SHE).

Fig. 11 delves into the changes in static read/write noise tolerance (RNM/WNM) of 6T-SRAM in BSPDN and FSPDN structures with and without SHE. It can be seen that under the influence of SHE, the static read/write noise tolerance of 6T-SRAM will experience indiscriminate drift, and the static read/write noise tolerance of SRAM with BSPDN and FSPDN structures shows different performance trends. The static read/write noise tolerance performance of BSPDN structure is better.

In general, the maximum allowed square length is used to evaluate the static noise margin. Fig. 12 shows the butterfly curves of the write noise margin (WNM) with and without self-heating effects, and correspondingly, the results of the read noise margin are also shown in the figure. WNM decayed from 320 mV (without self-heating) to 285 mV (with self-heating), while RNM degraded from 135 mV to 110.4 mV. The anti-interference ability of RNM is relatively weak. From the figure, it can also be seen that under high VDD stress, self-heating becomes more significant and exacerbates the degradation of circuit performance. When VDD is equal to 1.0 V, due to severe self-heating, RNM decreases by 29 mV (from 178 mV to 149 mV).

Fig. 13 shows a comparative evaluation of the electromigration (EM) lifetime of various materials by analyzing the effect of self-heating on temperature and using Black's empirical equation [21]. The self-heating experienced at the final output port (Nd) of the interconnection is the most

severe and requires more attention. Therefore, the evaluation is based on the worst-case scenario in BSM. According to the Black's empirical equation [22], the EM degradation of different materials [23] was predicted under the influence of self-heating, as shown in the figure. Compared with Cu, the EM lifetime of the interconnect metal Ni interconnect under the influence of self-heating has increased by 7.7 times.

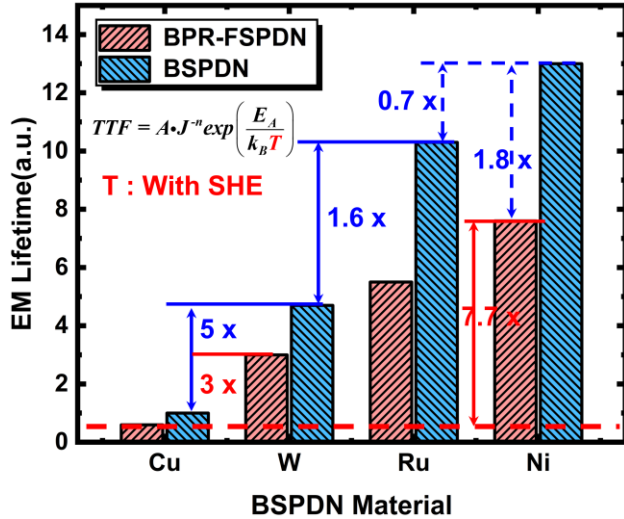


Figure 13. Using different advanced interconnect materials to predict temperature in SRAM and EM lifetime prediction in the worst PDN with SHE based on Black's empirical equation.

IV. CONCLUSIONS

This work is based on the Device-PDN-Circuit structure and analyzes the impact of self-heating on circuit performance and BSPDN structure interconnection reliability. By analyzing the temperature and heat flux distribution of different structures of FSPDN and BSPDN, the influence of self-heating effect on different structures and circuit structures was compared. The influence of device self-heating on circuit infrared voltage drop and temperature balance performance indicators was further discussed, providing guidance for the design of BSPDN.

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